from various hematopoietic cell lineages. Most dendritic cells are tissue bound, where they play an important role in bridging innate immunity and acquired immunity.

Dendritic cells can be classified by their location:

- Langerhans cells found in the skin and mucus membranes
- Interstitial dendritic cells which populate most organs (heart, lungs, liver, kidney, gastrointestinal tract)
- Interdigitating cells present in T cell areas of secondary lymphoid tissue and the thymic medulla.
- Circulating dendritic cells in the blood and lymph.

e. Lymphocytes

Lymphocytes are the major cells of the specific immunity. Lymphocytes can be divided into three populations: T cells, B cells, and NK (natural killer) cells. Clusters of differentiation are group of monoclonal antibodies that identify the same cell surface molecule. The cell surface molecule is designated CD (cluster of differentiation followed by a number (CD1, CD2).

i) B Lymphocytes

B lymphocytes mature within the bone marrow. When they leave bone marrow, each expresses a unique antigen binding receptor on its membrane. The B cell receptor is a membrane bound antibody molecule (Figure 13.13a). When a naive B cell, first encounters the antigen that matches its membrane bound antibody, the binding of the antigen to the antibody causes the cell to divide rapidly. Its progeny differentiate into memory B cells and effector B cells called plasma cells

Memory B cells have a longer life span than native cells. They express the same membrane bound antibody as their parent naive B cell. Plasma cells do not express membrane bound antibody. Plasma cells secrete large quantities of antibodies. Secreted antibodies are the major effector molecules of humoral immunity.

ii) T Lymphocytes

T lymphocytes also arise in the bone marrow. T cells then migrate to the thymus to mature. During its maturation within thymus, the T cells express a unique antigen binding molecule called the T cell receptor (Figure 13.13b) on its membrane. Unlike membrane bound antibodies on B cells, which can recognize antigen alone, T cell receptor can recognize only antigen that is bound to MHC molecules. There are two major types of MHC molecules. Class I MHC molecules are expressed by all nucleated cells. Class II MHC molecules are expressed only by antigen presenting cells. When a naive T cell encounters antigen combined with an MHC molecule on a cell the T cell proliferates and differentiates into memory T cell and various effector T cells.

There are two subpopulations of T cells: T helper (TH) and T cytotoxic (TC) cells. Although a third type of T cells called a T suppressor (T_S) cell, has been postulated, recent evidence suggests that it may not be distinct from the TH and TC subpopulations. T cells displaying CD4 function as TH cells whereas; those displaying CD8 function as TC cells (Figure 13.14).

After a TH cell recognizes and interacts with an antigen-MHC class II molecule complex, the cell is activated. It becomes an effector cell that secretes cytokines. The secreted cytokines activate B cells, TC cells,

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